

AGENT RUNTIME COST & RISK AUDIT REPORT

Generated: 2026-06-22 18:44:32 UTC | Behavioral Efficiency Score: 88/100
(Deterministic Baseline)

🔗 **Metric Calculation Formula:** Score equals 100 minus the Sum of Deductions. Penalty Weights: RetryLoop is -25, RepeatedRead is -15, RapidScan is -8, and ProtectedConfigMutation is -4.

Section 1 — Executive Summary

Audited session cost: \$0.831
Audited waste footprint: \$0.069 (8% of session cost)
Monthly projection: \$103.72 wasted at 50 sessions/day
(Assumption: 50 sessions/day * 30 days/month)
Critical alerts: 1 protected target modification alert
Action required: 2 items (see Technical Findings below)

Efficiency Benchmark: An optimized baseline execution of this sessi
of only \$0.762. Eliminating the identified be
patterns yields an immediate operational cost

Section 2 — Technical Findings

2.1 Budget & Efficiency Telemetry

Evaluates agent processing behavior against runaway API consumption and token waste patterns.

- **Runtime Health Status:** 🟡 REVIEW REQUIRED (High-Severity Incidents Tracked)

Audited Financial Metric	Value	Context
Session Cost (audited)	\$0.8308	Real-time computed API credit draw.
Audited Volumetric Waste	\$0.0691	8% of session runtime consumed by infractions.
Projected Monthly Waste	\$103.72	Scaled footprint at 50 developer sessions/day.

2.2 Behavioral Cost Indicators

Observed Waste Pattern	Target Count	Financial & Operational Implication
Estimated Wasteful Operations	4	Volumetric measure of redundant processing activity.

Retry Loop Incidents	0	Runaway compute waste; agent is stuck executing broken pipelines.
Repeated Target Reads	0	Context window waste; agent mismanaged previously read details.
Rapid Enumeration Bursts	1	Elevated context acquisition overhead from excessive repository scanning.
Protected Config Mutations	1	Critical corporate infrastructure configuration mutation risks.

2.3 Resource Allocation Ledger

- 📁 Context Gathering (READ): 8 file ingestion operations.
 - 📄 State Modifications (WRITE): 6 codebase mutation footprints.
 - ⚙️ Runtime Execution (BASH): 4 environment pipeline requests.
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2.4 Itemized Cost & Risk Breakdowns

[1] Excessive Repository Context Scan - RESOURCE_DRAIN (MEDIUM)

- **Impact Asset Target:** Multiple Files (main.py, models.py, routers/auth.py)
- **Triggering Samples:** 3 wasteful iterations inside time window constraint
- **Estimated Cost Attribution:** 10639 tokens allocated within window
- **Defensible Forensic Evidence:**
 - [11:14:36] READ main.py
 - [11:14:39] READ models.py
 - [11:14:41] READ routers/auth.py

[2] Protected Configuration Modification - COMPLIANCE_VIOLATION (HIGH)

- **Impact Asset Target:** .env
 - **Triggering Samples:** 1 wasteful iterations inside time window constraint
 - **Estimated Cost Attribution:** 12410 tokens allocated within window
 - **Defensible Forensic Evidence:**
 - [Target Match] Agent created/modified a credential file: .env
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2.5 Recommended Action Items

Recommendation #1

🔗📄 **Source Provenance Citation:** Linked to engine anomaly footprint detected by RapidFileEnumerationDetector on resource target Multiple Files (main.py, models.py, routers/auth.py).

1. Observed Behavior: The system detected an excessive repository context scan, evidenced by rapid sequential reads of multiple files including 'main.py', 'models.py',

and 'routers/auth.py' within a short time frame.

2. Potential Impact: Unnecessary file enumeration and ingestion can lead to increased operational costs due to higher token consumption, degraded performance from processing irrelevant data, and potential exposure of sensitive information if the accessed files contain data not pertinent to the task.
3. Suggested Review Action: Review the application's context management strategy to ensure that only necessary files and directories are made available to the language model for processing.

Suggested Strategic Review:

Implement explicit context management policies, such as defining task-specific scopes for file access or configuring directory exclusion rules, to limit the LLM's operational context strictly to relevant information. Alternatively, refine prompt design to explicitly guide the model to only access and process specified files or data segments.

Recommendation #2

🚩 **Source Provenance Citation:** Linked to engine anomaly footprint detected by `SecretAccessDetector` on resource target `.env`.

1. Observed Behavior: An agent was observed creating or modifying a credential file named `.env`, which is designated as a protected configuration.
2. Potential Impact: Unauthorized modification of sensitive configuration files can lead to exposure of credentials, system misconfiguration, or service disruption, potentially compromising data integrity and security.
3. Suggested Review Action: Review the operational scope and access permissions of the agent responsible for the modification. Implement stricter context management and directory exclusion rules to prevent unintended interaction with sensitive configuration files.

Suggested Strategic Review:

Implement robust context management and strict directory exclusion rules to prevent agents from accessing or modifying sensitive configuration files. Ensure task-specific scopes are narrowly defined to limit operational reach to only necessary directories and file types.